

Lesson 3 · Sketching Gradient Functions

NCEA Level 2 Mathematics · AS 91262 Apply calculus methods in solving problems · Handout

1 The one idea

The **height** of the f' graph is the **steepness** of the f graph.

Where f is flat (a turning point), f' is on the x -axis. Where f goes up, f' is above the axis; where f goes down, f' is below.

Always draw the new graph **directly under** the given one, using the same x -scale, and line up the key x -values with a ruler.

2 Translating features

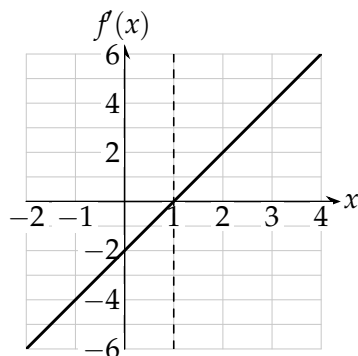
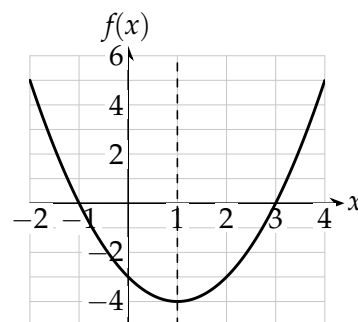
on the graph of f	on the graph of f'
turning point (maximum or minimum)	crosses or touches the x-axis ($f' = 0$)
increasing (going up)	above the x -axis ($f' > 0$)
decreasing (going down)	below the x -axis ($f' < 0$)
steepest point (point of inflection)	turning point of f'
straight line with gradient m	horizontal line at height m
parabola	straight line
cubic	parabola
quartic	cubic

The power drops by one: that is why a parabola gives a line and a cubic gives a parabola.

3 Lines and parabolas

A straight line has the same gradient everywhere, so f' is a **horizontal line**. E.g. $f(x) = 2x + 1$ gives $f'(x) = 2$ for every x .

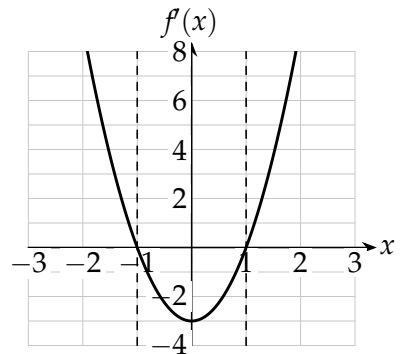
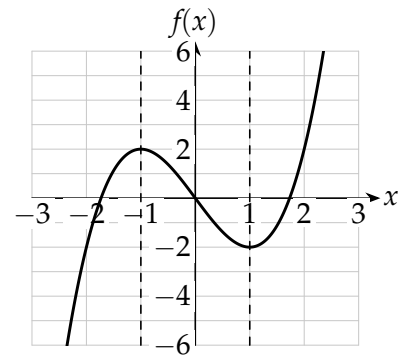
A parabola has one turning point, so f' is a **straight line through that x -value**. For the parabola on the right (minimum at $x = 1$): down then up, so f' goes from below the axis to above it, crossing at $x = 1$. Here $f(x) = x^2 - 2x - 3$ and $f'(x) = 2x - 2$.



4 Cubics and quartics

Write it in this order

1. Mark the x -value of each turning point of f on the new axis: these are the **zeros of f'** .
2. In each interval: f up $\rightarrow f'$ above; f down $\rightarrow f'$ below.
3. The steepest point of f gives the turning point of f' .
4. Join smoothly. Check the type: cubic $\rightarrow$ parabola.



Worked example 1

$f(x) = x^3 - 3x$ has a maximum at $x = -1$ and a minimum at $x = 1$.

Step 1 Zeros of f' at $x = -1$ and $x = 1$.

Step 2 Up, down, up: f' is above, below, above.

Step 3 Steepest downhill at $x = 0$: lowest point of f' .

So f' is an up-parabola: $f'(x) = 3x^2 - 3$.

A quartic with three turning points gives a cubic f' with three zeros.

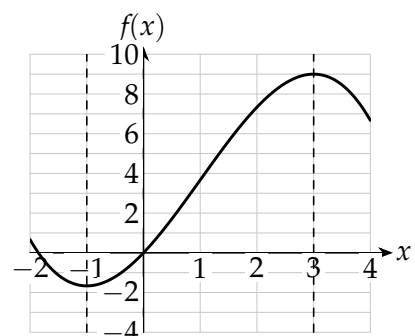
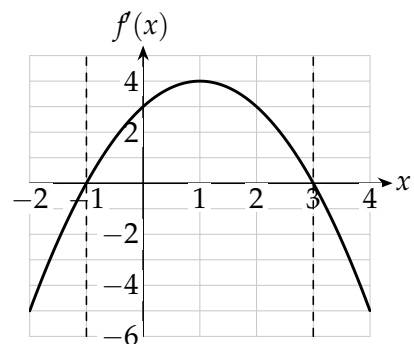
5 From f' back to f

f' graph	f graph
crosses from + to -	maximum
crosses from - to +	minimum
above the axis	increasing
below the axis	decreasing
turning point of f'	steepest point

The f' graph shows gradients only, so the **height of f is unknown**: sketch a possible graph. Any vertical shift is correct.

Worked example 2

f' (top right) crosses from - to + at $x = -1$ and from + to - at $x = 3$. So f has a **minimum at $x = -1$** and a **maximum at $x = 3$** ; it is a negative cubic. One possible f is drawn under it.



6 Versions that get no credit

sketch	why it fails
f' with the same humps as f	turning points of f must become zeros of f'
f' of a parabola drawn curved	a parabola's gradient changes at a constant rate: a line
f drawn through the zeros of f'	zeros of f' are the turning points of f
key points not lined up	the marker checks the x -values against the given graph

7 What each grade looks like here

Achieved	a correct sketch of f' for a line or a parabola, zeros in the right places.
Merit	a correct f' for a cubic or quartic (zeros, signs and shape), or a correct f from f' .
Excellence	a full explanation linking the two graphs, e.g. why a sharp corner has no gradient, or why the maximum of f' is not a maximum of f .

8 Checking with the fx-9750GIII

SET UP (SHIFT MENU) → Derivative: On. In GRAPH, TRACE then shows the gradient $\frac{dy}{dx}$ at each point. Trace along f : the sign of the gradient tells you which side of the axis f' is on.

9 Ways to lose marks

From the material plus teaching observation.

- L1 Copying the shape: the turning points of f become **x -intercepts** of f' , not turning points of f' .
- L2 Wrong side of the axis: where f goes down, f' is **below** the x -axis.
- L3 Wrong type of graph: a parabola's f' is a straight line; a cubic's f' is a parabola.
- L4 The steepest point of f not matched: it gives the turning point of f' .
- L5 Reverse sketch: the zeros of f' are the turning points of f , not where f crosses the axis.
- L6 Key x -values not lined up with the given graph, or the new graph not labelled.